Number, Surds, Bounds & Set Notation

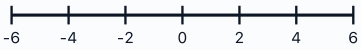
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Number, Surds, Bounds & Set Notation Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 6x + 20$ in the domain of Number, Surds, Bounds & Set Theory. By completing the square $f_1(x) = (x - h)^2 + k$, find the sum $k + f_1(5)$.



A 26

B 29

C 15

D 24

2 MCQ

A mathematical model in Number, Surds, Bounds & Set Theory is defined by the cubic function $y_2(x) = 2x^3 - 3x^2 + 15x - 8$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 69

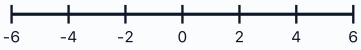
B 51

C 36

D 27

3 MCQ

In Number, Surds, Bounds & Set Theory, two displacement vectors are given by $\mathbf{u}_3 = \langle 3, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.



A -29

B 73

C 44

D 156

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4 MCQ

The linear growth profile shown in the diagram for Number, Surds, Bounds & Set Theory follows an arithmetic progression with first term $a_1 = 4$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 88

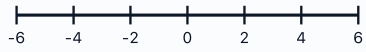
B 780

C 1320

D 690

5 MCQ

In Number, Surds, Bounds & Set Theory, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 12x)\,\mathrm{d}x$.



A 72

B 96

C 84

D 37

6 MCQ

In Number, Surds, Bounds & Set Theory, consider the quadratic function $f_6(x) = x^2 - 8x + 99$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(11)$.

A 219

B 215

C 132

D 208

7 MCQ

The graph illustrates the cubic model $y_7(x) = 2x^3 - 4x^2 + 33x - 15$ studied in Number, Surds, Bounds & Set Theory. Find the instantaneous gradient $\frac{\mathrm{dy}}{\mathrm{dx}}$ of the tangent to the curve at $x = 8$.

A 417

B 320

C 353

D 73

8 MCQ

In Number, Surds, Bounds & Set Theory, two displacement vectors are given by $\mathbf{u}_8 = \langle 4, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A -3

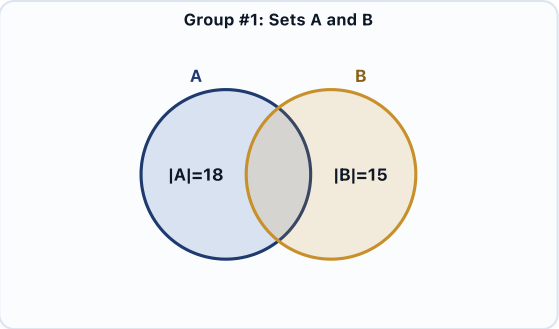
B 65

C 132

D 62

9 MCQ

In a survey of students in Year 11 Group #1, $|A| = 18$ study French, $|B| = 15$ study Spanish, $|A \cap B| = 5$ study both languages, and 4 study neither. Find the total number of students $|E|$.



A 37

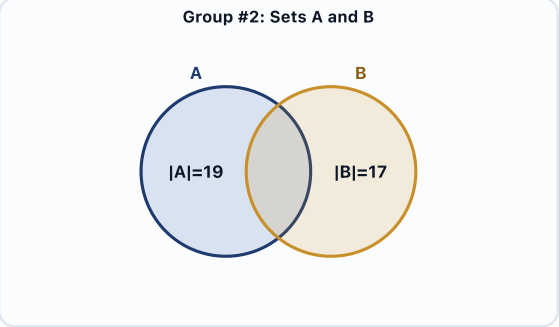
B 14

C 28

D 32

10 MCQ

In a survey of students in Year 11 Group #2, $|A| = 19$ study French, $|B| = 17$ study Spanish, $|A \cap B| = 6$ study both languages, and 5 study neither. Find the total number of students $|E|$.



A 14

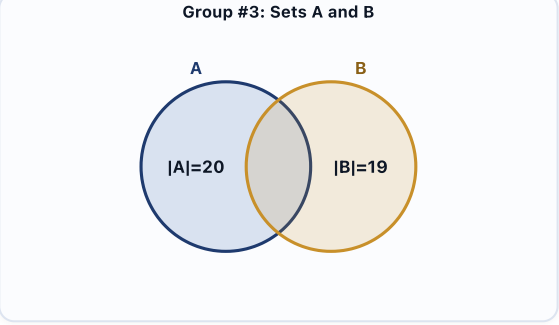
B 35

C 41

D 30

11 MCQ

In a survey of students in Year 11 Group #3, $|A| = 20$ study French, $|B| = 19$ study Spanish, $|A \cap B| = 7$ study both languages, and 6 study neither. Find the total number of students $|E|$.



A 32

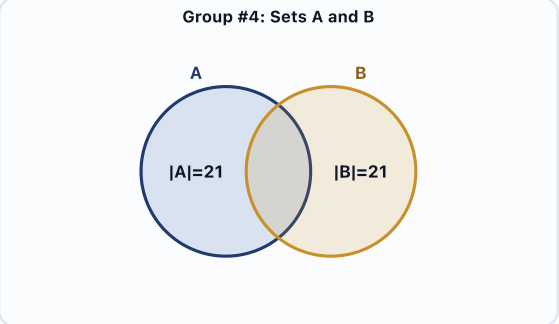
B 38

C 45

D 14

12 MCQ

In a survey of students in Year 11 Group #4, $|A| = 21$ study French, $|B| = 21$ study Spanish, $|A \cap B| = 8$ study both languages, and 7 study neither. Find the total number of students $|E|$.



A 49

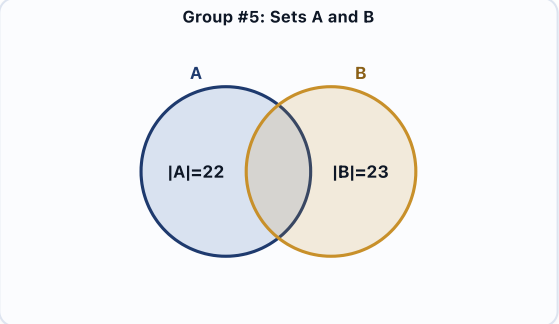
B 14

C 41

D 34

13 MCQ

In a survey of students in Year 11 Group #5, $|A| = 22$ study French, $|B| = 23$ study Spanish, $|A \cap B| = 9$ study both languages, and 8 study neither. Find the total number of students $|E|$.



A 44

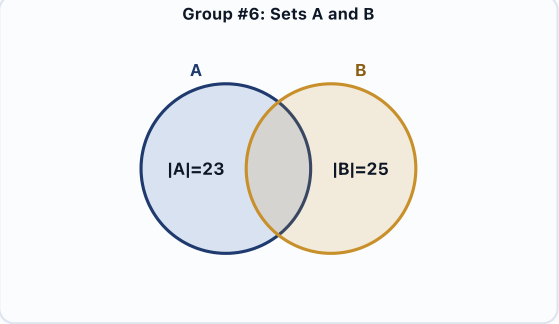
B 36

C 53

D 14

14 MCQ

In a survey of students in Year 11 Group #6, $|A| = 23$ study French, $|B| = 25$ study Spanish, $|A \cap B| = 10$ study both languages, and 9 study neither. Find the total number of students $|E|$.



A 14

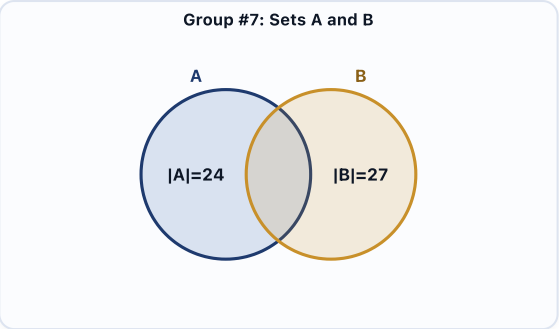
B 57

C 47

D 38

15 MCQ

In a survey of students in Year 11 Group #7, $|A| = 24$ study French, $|B| = 27$ study Spanish, $|A \cap B| = 11$ study both languages, and 10 study neither. Find the total number of students $|E|$.



A 50

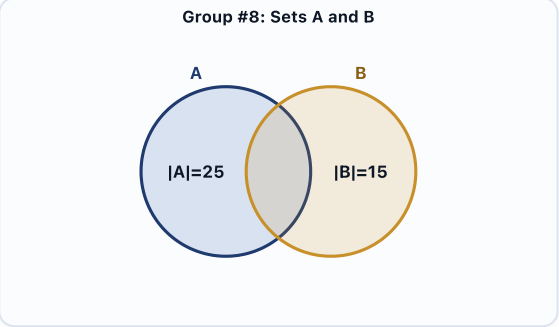
B 61

C 40

D 14

16 MCQ

In a survey of students in Year 11 Group #8, $|A| = 25$ study French, $|B| = 15$ study Spanish, $|A \cap B| = 12$ study both languages, and 4 study neither. Find the total number of students $|E|$.



A 44

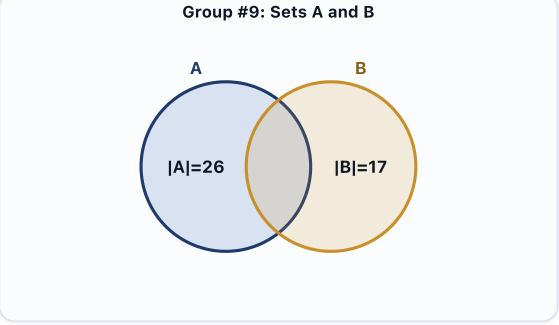
B 28

C 32

D 14

17 MCQ

In a survey of students in Year 11 Group #9, $|A| = 26$ study French, $|B| = 17$ study Spanish, $|A \cap B| = 5$ study both languages, and 5 study neither. Find the total number of students $|E|$.



A 48

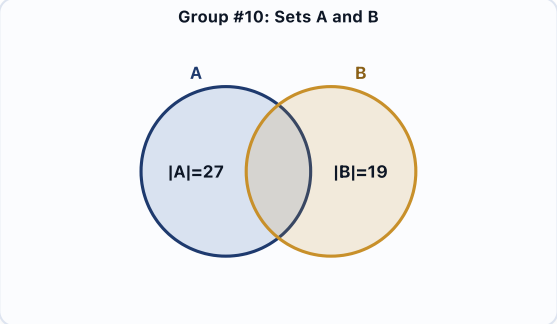
B 38

C 14

D 43

18 MCQ

In a survey of students in Year 11 Group #10, $|A| = 27$ study French, $|B| = 19$ study Spanish, $|A \cap B| = 6$ study both languages, and 6 study neither. Find the total number of students $|E|$.



A 14

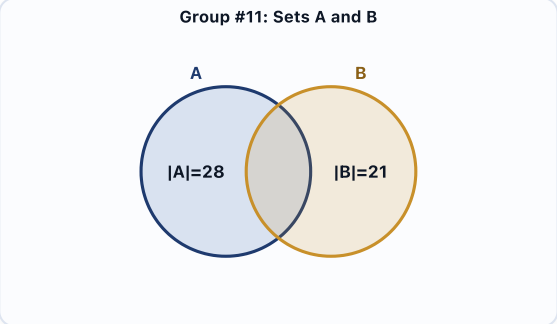
B 46

C 40

D 52

19 MCQ

In a survey of students in Year 11 Group #11, $|A| = 28$ study French, $|B| = 21$ study Spanish, $|A \cap B| = 7$ study both languages, and 7 study neither. Find the total number of students $|E|$.



A 42

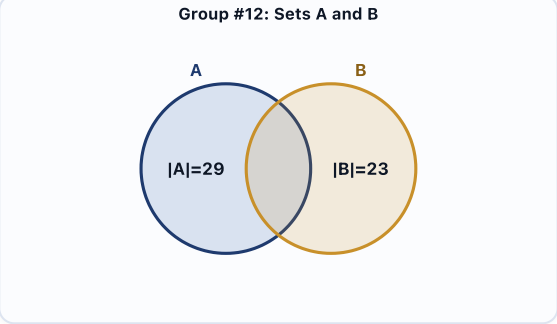
B 56

C 49

D 14

20 MCQ

In a survey of students in Year 11 Group #12, $|A| = 29$ study French, $|B| = 23$ study Spanish, $|A \cap B| = 8$ study both languages, and 8 study neither. Find the total number of students $|E|$.



A 60

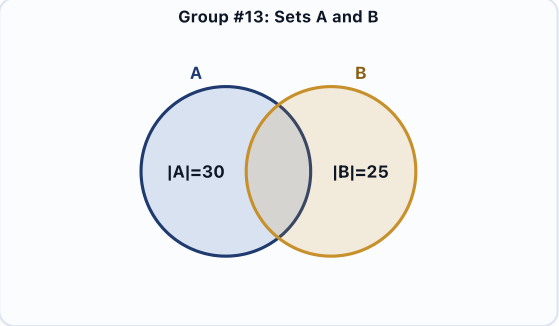
B 52

C 44

D 14

21 MCQ

In a survey of students in Year 11 Group #13, $|A| = 30$ study French, $|B| = 25$ study Spanish, $|A \cap B| = 9$ study both languages, and 9 study neither. Find the total number of students $|E|$.



A 64

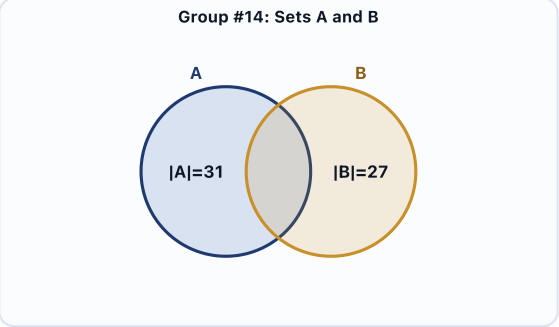
B 14

C 55

D 46

22 MCQ

In a survey of students in Year 11 Group #14, $|A| = 31$ study French, $|B| = 27$ study Spanish, $|A \cap B| = 10$ study both languages, and 10 study neither. Find the total number of students $|E|$.



A 48

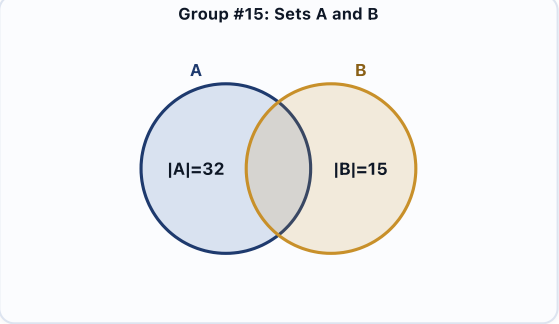
B 14

C 58

D 68

23 MCQ

In a survey of students in Year 11 Group #15, $|A| = 32$ study French, $|B| = 15$ study Spanish, $|A \cap B| = 11$ study both languages, and 4 study neither. Find the total number of students $|E|$.



A 51

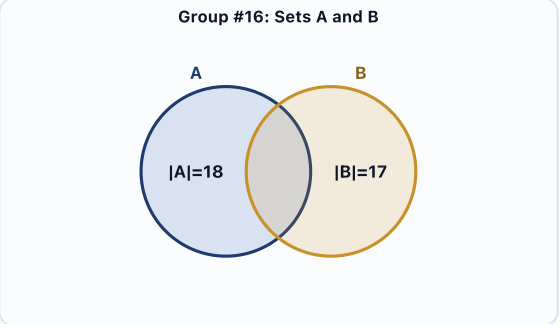
B 14

C 40

D 36

24 MCQ

In a survey of students in Year 11 Group #16, $|A| = 18$ study French, $|B| = 17$ study Spanish, $|A \cap B| = 12$ study both languages, and 5 study neither. Find the total number of students $|E|$.



A 14

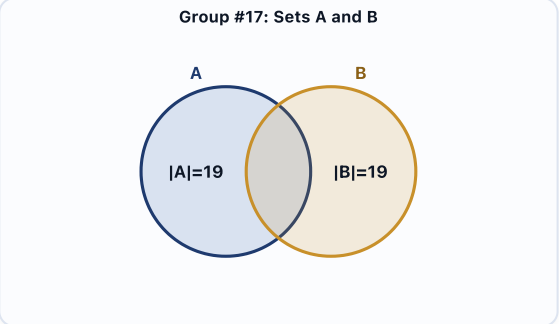
B 28

C 23

D 40

25 MCQ

In a survey of students in Year 11 Group #17, $|A| = 19$ study French, $|B| = 19$ study Spanish, $|A \cap B| = 5$ study both languages, and 6 study neither. Find the total number of students $|E|$.



A 39

B 44

C 14

D 33